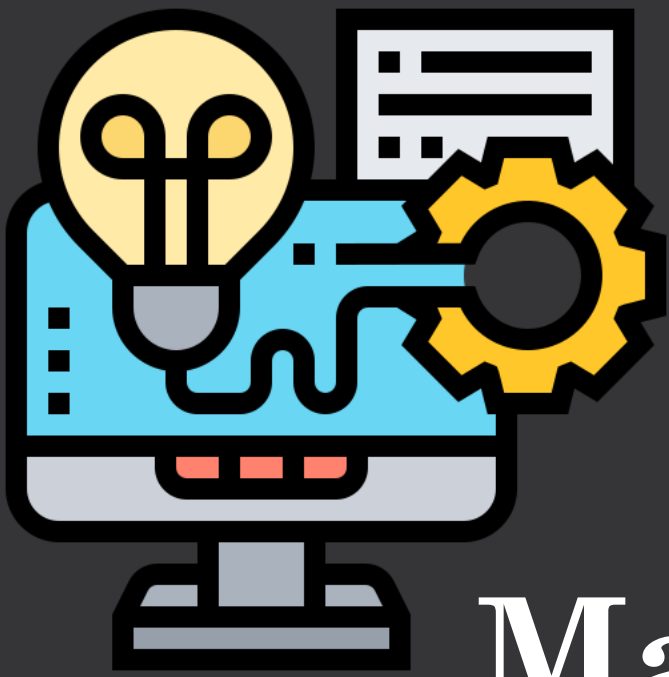




Machine Learning

Lecture 01

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RA: Imran Nawar

28 September 2025

1

Overview

- Traditional Programming vs. Machine Learning
- **Mathematics for Machine Learning**
 - Linear Algebra
 - Calculus
 - Statistics and Probability
- **Machine Learning**
 - Introduction to Machine Learning
 - Machine Learning Examples
 - Evolution of Machine Learning
 - Types of Machine Learning
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning
 - Workflow of Machine Learning
- **Key Concepts**
 - Data and Features
 - Algorithms and Models
 - Training and Testing
 - Evaluation Metrics
- Common Algorithms
- Tools and Frameworks
- Challenges and Limitations
- Future Trends

Traditional Programming vs Machine Learning

Traditional Programming

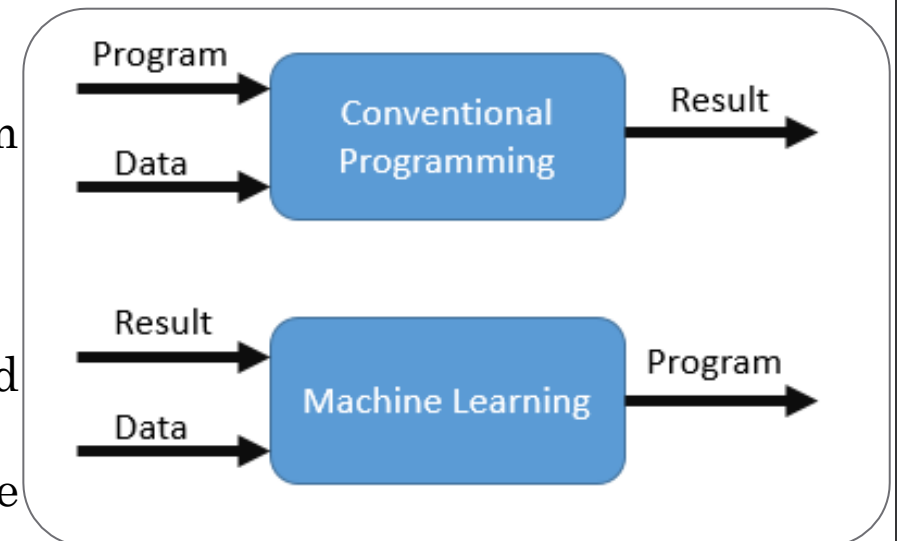
- **Approach:** Write explicit rules and logic for specific tasks.
- **Input:** Rules (logic) + **Data** → **Output**.
- **Key Focus:** Programmer defines all possible scenarios.
- **Example:**
 - Use if-else statements to classify emails as spam or not.

Machine Learning

- **Approach:** Learn patterns from data to make predictions or decisions.
- **Input:** **Data** + **Output** (labels) → Learns **Rules** (Model).
- **Key Focus:** Relies on high-quality training data.
- **Example:**
 - Train a spam classifier with labeled examples of spam and non-spam emails.

Key Difference

- **Traditional Programming:** Fixed rules; limited flexibility.
- **Machine Learning:** Adaptive rules; improves with more data.



Mathematics for Machine Learning

Mathematics: The Foundation of Machine Learning

Why Mathematics Matters in ML

- Machine learning algorithms are built on mathematical foundations translated into code.
- Understanding key concepts helps you grasp how algorithms learn and why they behave the way they do.
- Essential for optimizing and debugging ML systems.
- Foundation for reading research papers and documentation.

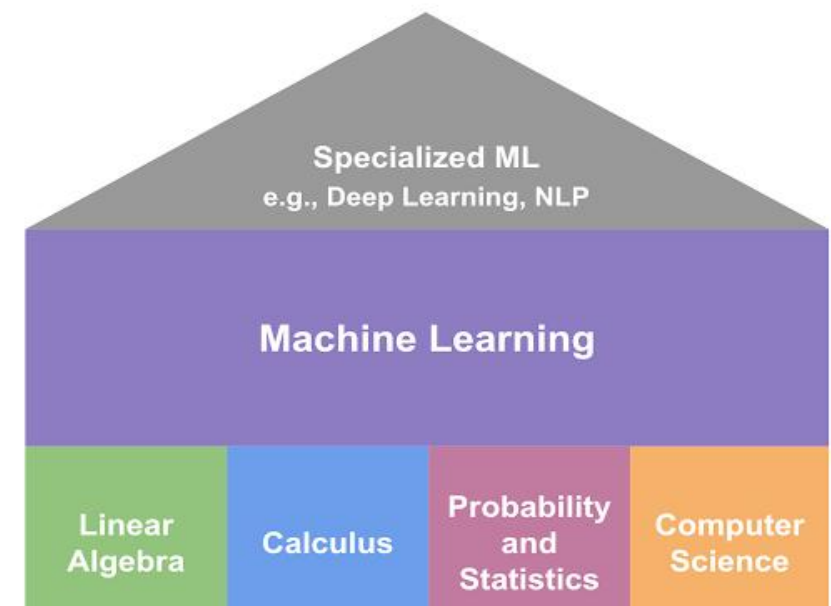
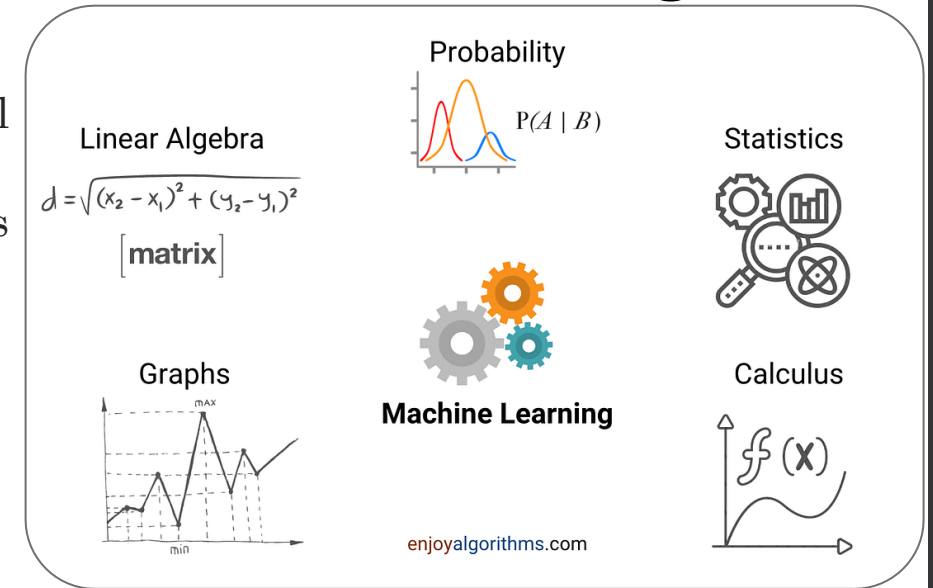
Key Areas for Mathematics in ML

- **Linear Algebra:** Vectors, matrices, transformations are foundational for data representation and manipulation.
- **Probability & Statistics:** Essential for model evaluation, hypothesis testing, and analyzing uncertainty in data.
- **Calculus:** Derivatives, gradients and optimization techniques are central to training models (e.g., gradient descent).

If you want an ML career:

- Machine Learning Engineer
- Data Scientist
- AI Engineer
- Data Analyst (for analytical and statistical modeling tasks)

You should focus on the mathematic concepts described here.



Linear Algebra: A Core Pillar of Machine Learning

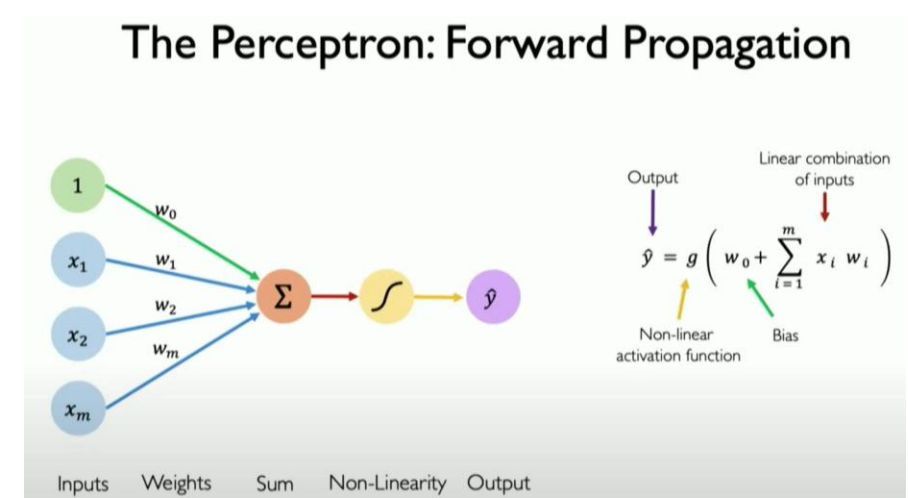
- Linear algebra is a branch of mathematics that deals with vectors, matrices, and how they transform or interact.

Key Concepts to Know:

- Vectors & operations:** For example, adding vectors, multiplying by a number.
- Systems of linear equations:** Solving multiple equations at once.
- Matrix operations:** Like addition, multiplication, transpose.
- Linear transformations:** Changing vectors while keeping their structure.

Why It Matters in Machine Learning:

- Linear algebra is used to:
 - Represent data (as vectors and matrices).
 - Power key algorithms (e.g., **PCA**, **SVD**).
 - Enable neural network computations (matrix operations in each layer).
 - Perform dimensionality reduction (simplifying complex data).



Scalar	Vector(s)	
1	$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$
Matrix	Tensor	
$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$	$\begin{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \end{bmatrix}$	$\begin{bmatrix} \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix} \end{bmatrix}$

Essential Linear Algebra Concepts for ML

- **Matrix:** Rectangular array of numbers ($m \times n$ dimensions).
 - **Matrix Elements:** Entries of matrix.
 - **Operations:** Addition (same dimensions), scalar multiplication, matrix multiplication.
- **Vector:** Special case of matrix ($n \times 1$ or $1 \times n$).
 - Represents data points, features, or model parameters.
- **Identity Matrix (I):** Square matrix with 1s on diagonal, 0s elsewhere.
- **Matrix Transpose (A^T):** Rows become columns and vice versa
 - Essential for many ML computations

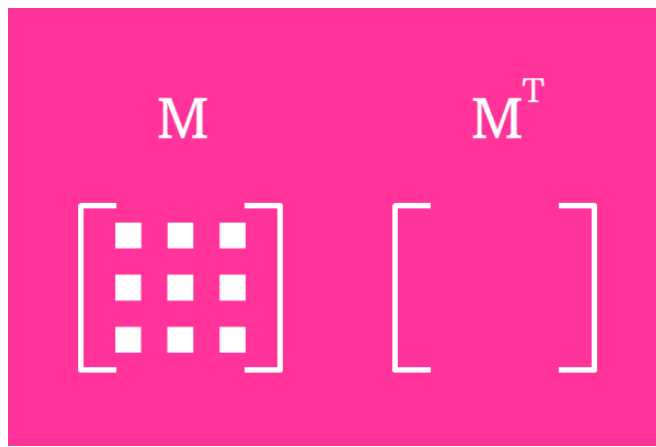
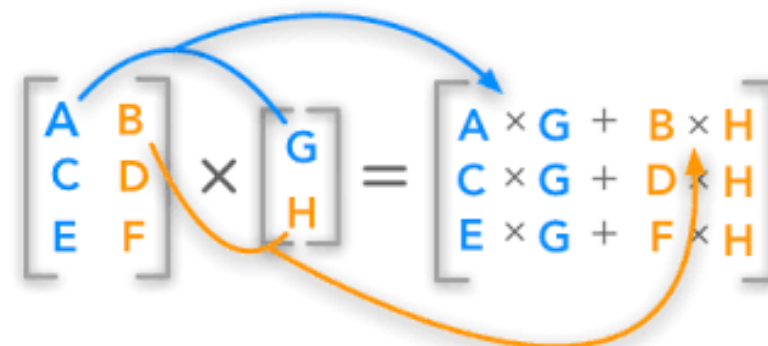
Formula:

For matrices A and B:

$$C_{i,j} = A_{i,j} + B_{i,j}$$

For matrices A and scalar k:

$$(kA)_{i,j} = k \cdot A_{i,j}$$



$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2 x 2

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3 x 3

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

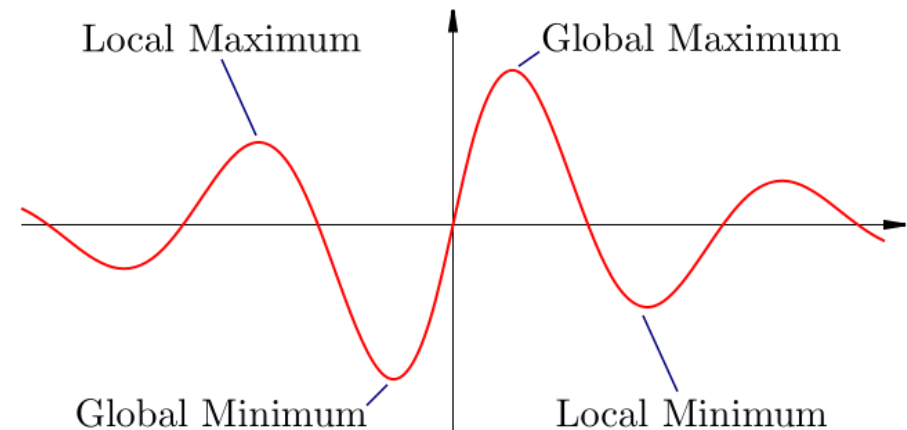
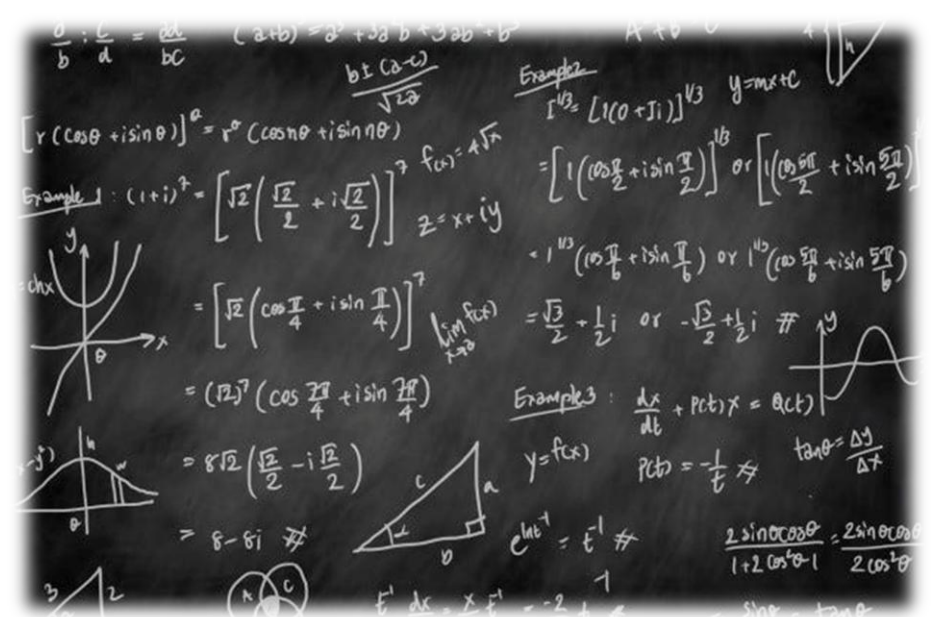
4 x 4

Calculus – The Engine of Machine Learning Optimization

- **Calculus** is the mathematics of change and rates. It is essential for understanding how ML models learn and improve through optimization.

Core Concepts for Machine Learning:

- **Derivatives:** Measure the rate of change of a function
- **Partial derivatives:** First-order derivatives w.r.t. one variable (others held constant).
- **Gradient:** A vector of all partial derivatives; direction of steepest ascent.
- **Second-Order Derivatives:** Describe how the slope itself changes (curvature).
- **Hessian Matrix:** Matrix of second-order partial derivatives; used to analyze curvature of loss surfaces.
- **Chain Rule:** Crucial for computing gradients in neural networks (backpropagation).



Statistics and Probability – Understanding Data and Uncertainty

Statistics in Machine Learning

- Statistics is the science of collecting, analyzing, summarizing, and interpreting data.
- In machine learning, it helps us make sense of large datasets and draw meaningful conclusions.
- **Why is Statistics Important in ML?**
 - **Summarizing Data:** Measures like *mean*, *median*, and *mode* help us understand the central tendency of data.
 - **Understanding Variability:** *Variance* and *standard deviation* show how much data points spread out from the average.
 - **Making Inferences:** Inferential statistics (e.g., *hypothesis testing* and *confidence intervals*) help us generalize from samples to populations, which is vital when evaluating model performance.
- **Applications in ML:**
 - Statistical concepts are used in feature selection, model evaluation, and understanding data distributions.

Mean

Add all the numbers then divide by the amount of numbers

9, 3, 1, 8, 3, 6

$$9 + 3 + 1 + 8 + 3 + 6 = 30$$

$$30 \div 6 = 5$$

The mean is 5

Median

Order the set of numbers, the median is the middle number

9, 3, 1, 8, 3, 6

1, 3, 3, 6, 8, 9

The median is 4.5

Mode

The most common number

9, 3, 1, 8, 3, 6

The mode is 3

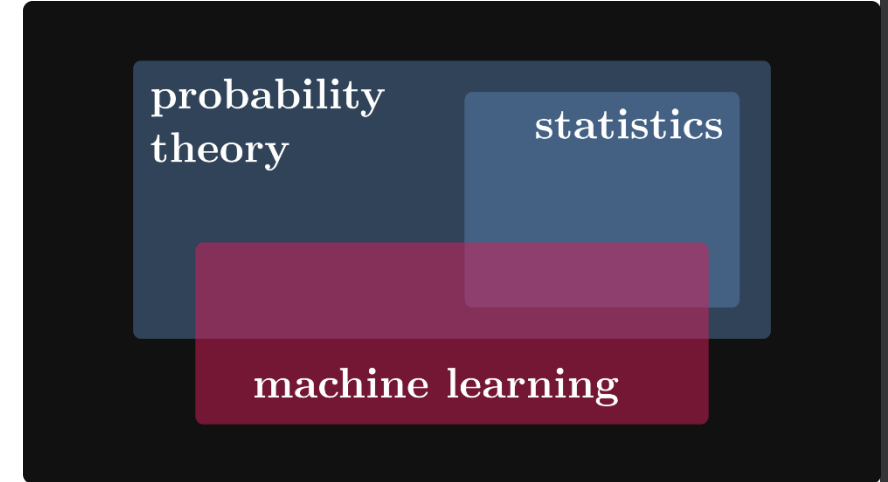
INFERENCE STATISTICS



Statistics and Probability – Understanding Data and Uncertainty

Probability in Machine Learning

- Probability measures how likely an event is to happen, ranging from 0 (impossible) to 1 (certain).
- **Why It Matters in ML?**
 - **Handling Uncertainty:** Helps models deal with uncertain or noisy data (e.g., spam detection).
 - **Probability Distributions:** Many algorithms assume data follows certain distributions (like *normal distribution*), which helps in model design and evaluation.
 - **Bayesian Inference:** Updates predictions as new data arrives, foundational for algorithms like Naive Bayes.
- **Applications in ML:**
 - Algorithms such as Naive Bayes, logistic regression, and Markov models are built on probability theory.
 - Probability helps quantify how confident we are in model predictions and guides decision-making in uncertain situations.



Machine Learning

Introduction To Machine Learning

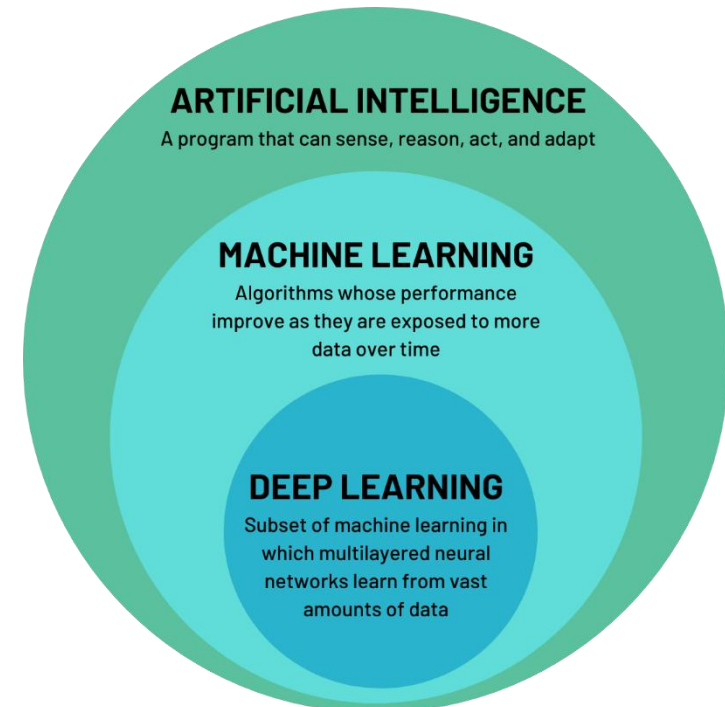
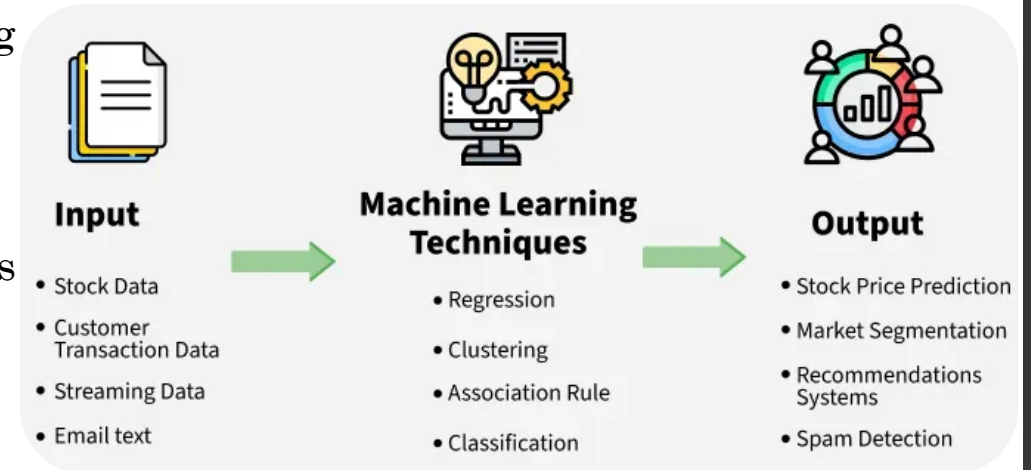
- Machine learning is the science (and art) of programming computers so they can learn from data.
- [Machine learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

— Arthur Samuel, 1959

- “A computer program is said to *learn* from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

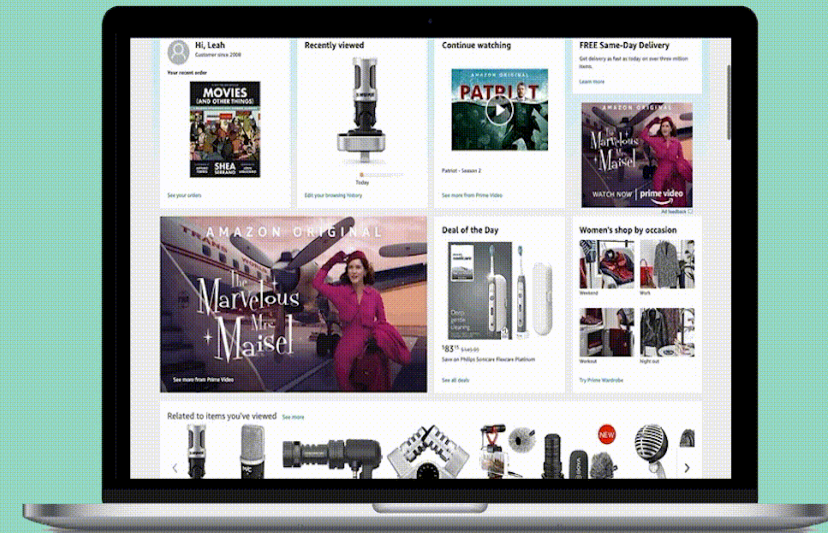
—Tom Mitchell, 1997

- [Machine Learning means a computer improves at a task (T) over time as it gains experience (E), measured by a performance metric (P)].



ML Examples

- **Complex Tasks Too Difficult to Program**
 - **Autonomous Vehicles:** Navigate roads, detect obstacles, make real-time decisions.
 - **Computer Vision:** Medical imaging (tumor detection), face recognition.
 - **Natural Language Processing:** Language translation, chatbots, sentiment analysis.
- **Intelligent Personalization Systems**
 - **Recommendation Engines:** Amazon product suggestions.
- **Pattern Discovery in Big Data**
 - **Fraud Detection:** Real-time suspicious transaction identification.
 - **Predictive Analytics:** Equipment failure prediction, market trend analysis.
- **Why ML Excels:**
 - Traditional programming can not handle the complexity, variability, and scale
 - ML learns patterns from data and improves with experience.



Evolution of Machine Learning

Early Foundations (1950s – 1980s)

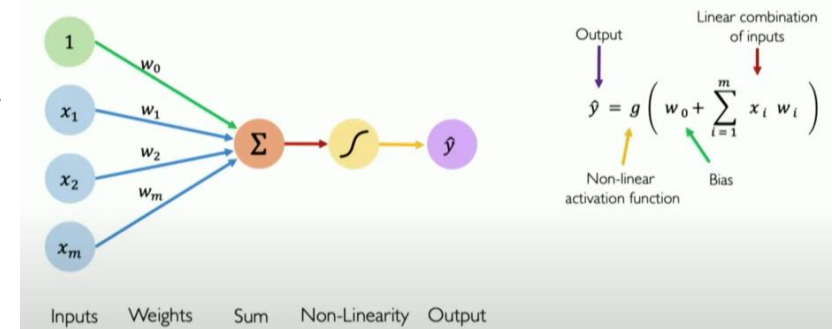
- **Dartmouth Conference (1956):**
 - Dartmouth Summer Research Project on Artificial Intelligence – officially founded AI as a field.
 - Researchers discussed creating intelligent machines.
- **Initial Algorithms and Theories:**
 - **Perceptron (1957):** *Frank Rosenblatt* – first trainable neural network.
 - **Decision Trees (1960s):** ID3 algorithm.
 - **Bayesian Networks (1980s):** Probabilistic reasoning models.

Key Algorithmic Breakthroughs (1980s – 1990s)

- **Neural Network Revival:**
 - Backpropagation (1986) – *Rumelhart, Hinton, and Williams* – enabled training deep networks
 - Support Vector Machines (1990s) – *Vapnik and Cortes* – powerful classification method



The Perceptron: Forward Propagation



Evolution of Machine Learning

Big Data Era (2000s-2010s):

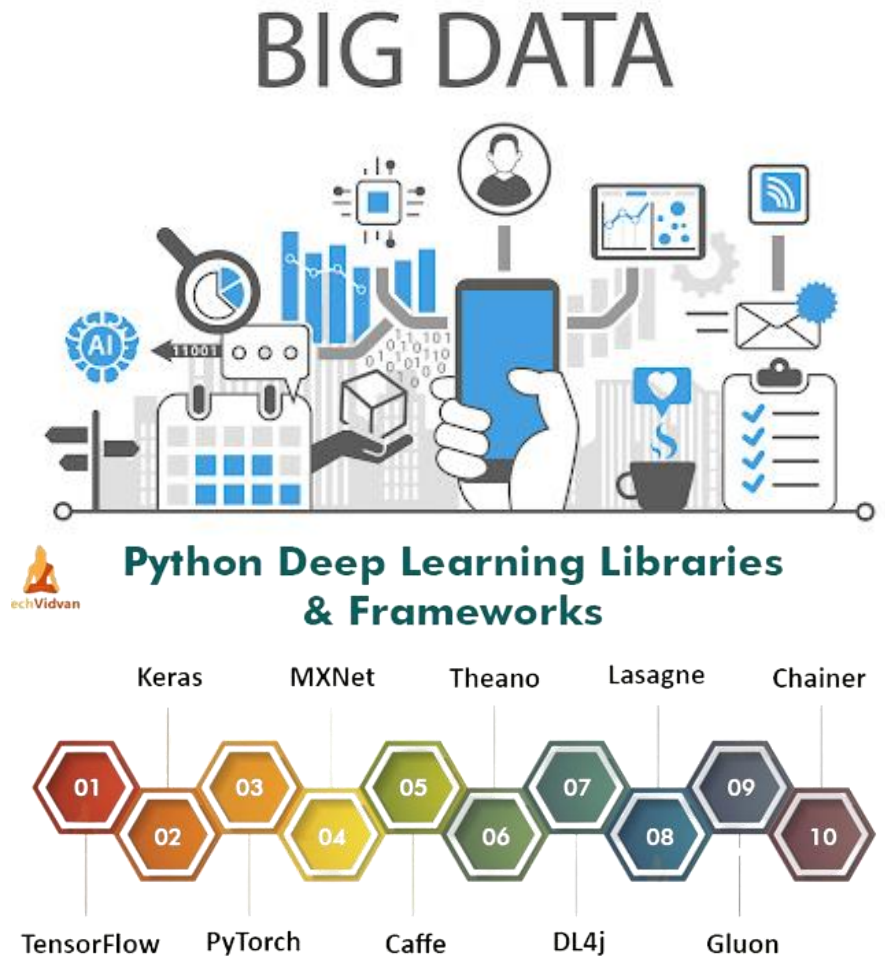
- **Rise of Computational Power:**
 - Internet and sensors generated massive datasets for training.
 - GPU acceleration revolutionized neural network training.
 - Open-source frameworks: TensorFlow, PyTorch, Scikit-learn.

Deep Learning Revolution (2010s-Present):

- **Breakthrough Architectures:**
 - Convolutional Neural Networks (CNNs) – image recognition
 - Recurrent Neural Networks (RNNs) – sequential data processing
- **Major Milestones:**
 - AlphaGo (2016) – Google DeepMind defeats world Go champion.
 - GPT Models (2018-present) – OpenAI's language models.
 - Transformer Architecture (2017) – “Attention is All You Need”

Current Era: Foundation Models

- Large-scale pre-trained models (GPT, BERT, DALL-E) transforming AI applications.



4 – 1



Types of Machine Learning

1. Supervised Learning

- Supervised Learning involves training a model on a labeled dataset, where the input data is paired with the correct output.
- “right answers” given.
- Supervised learning learn from data labeled with the “right answers”.

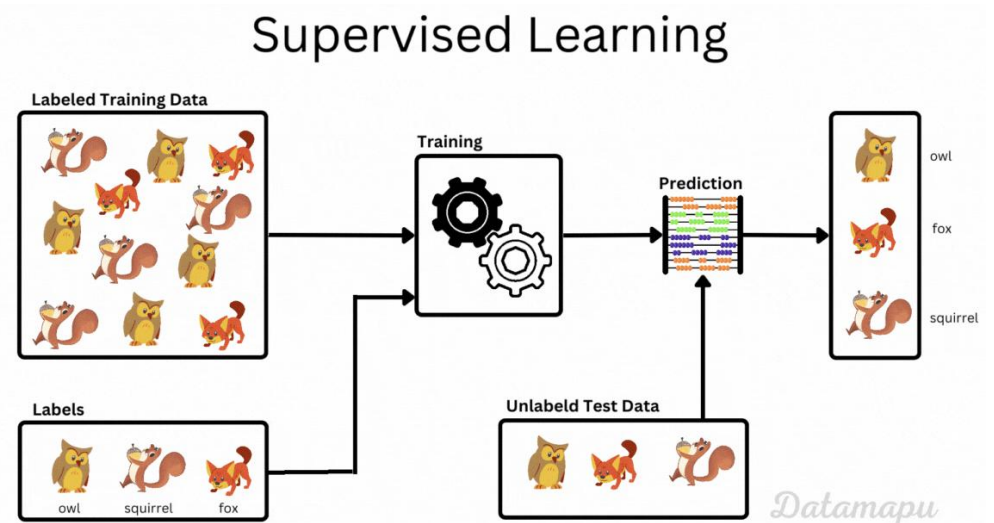
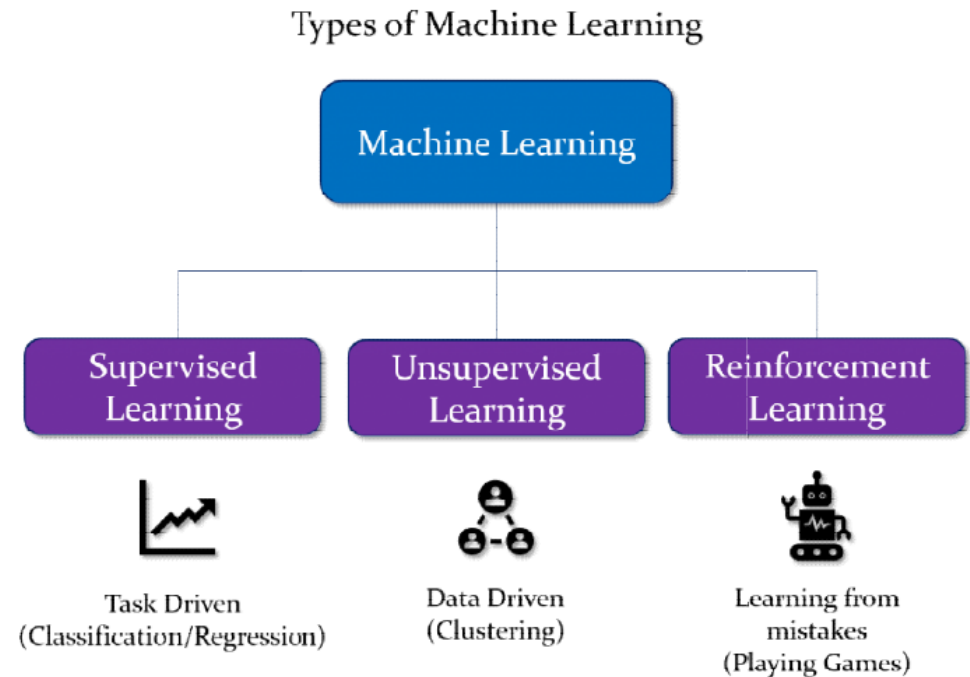
Common Tasks in Supervised Learning:

1. Regression:

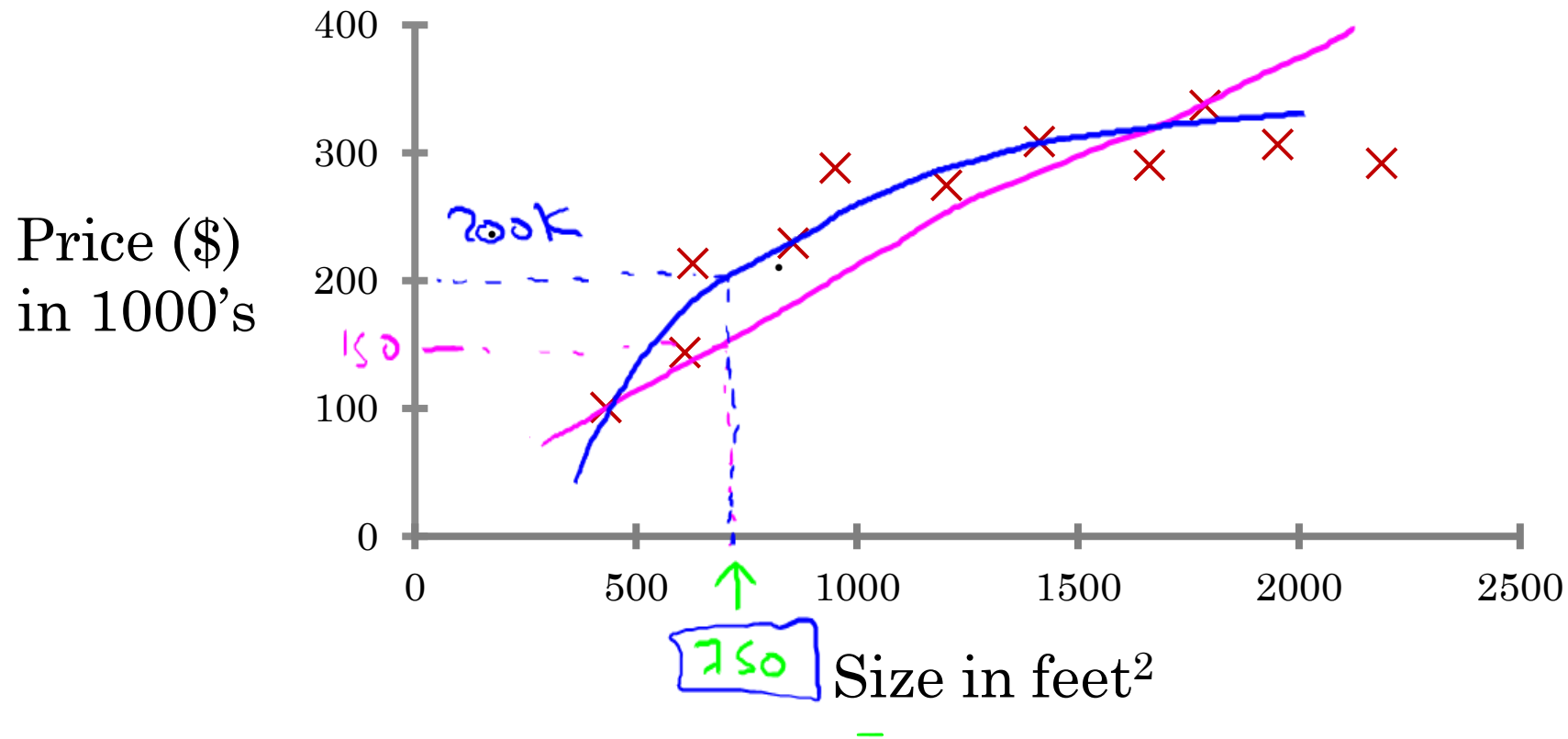
- Predict continuous valued output.
- For example, predicting house prices based on features like size, location, and age.

2. Classification:

- Predict discrete valued output.
- Categorizing input data into predefined classes.
- For example, identifying whether an email is spam or not spam based on its content.



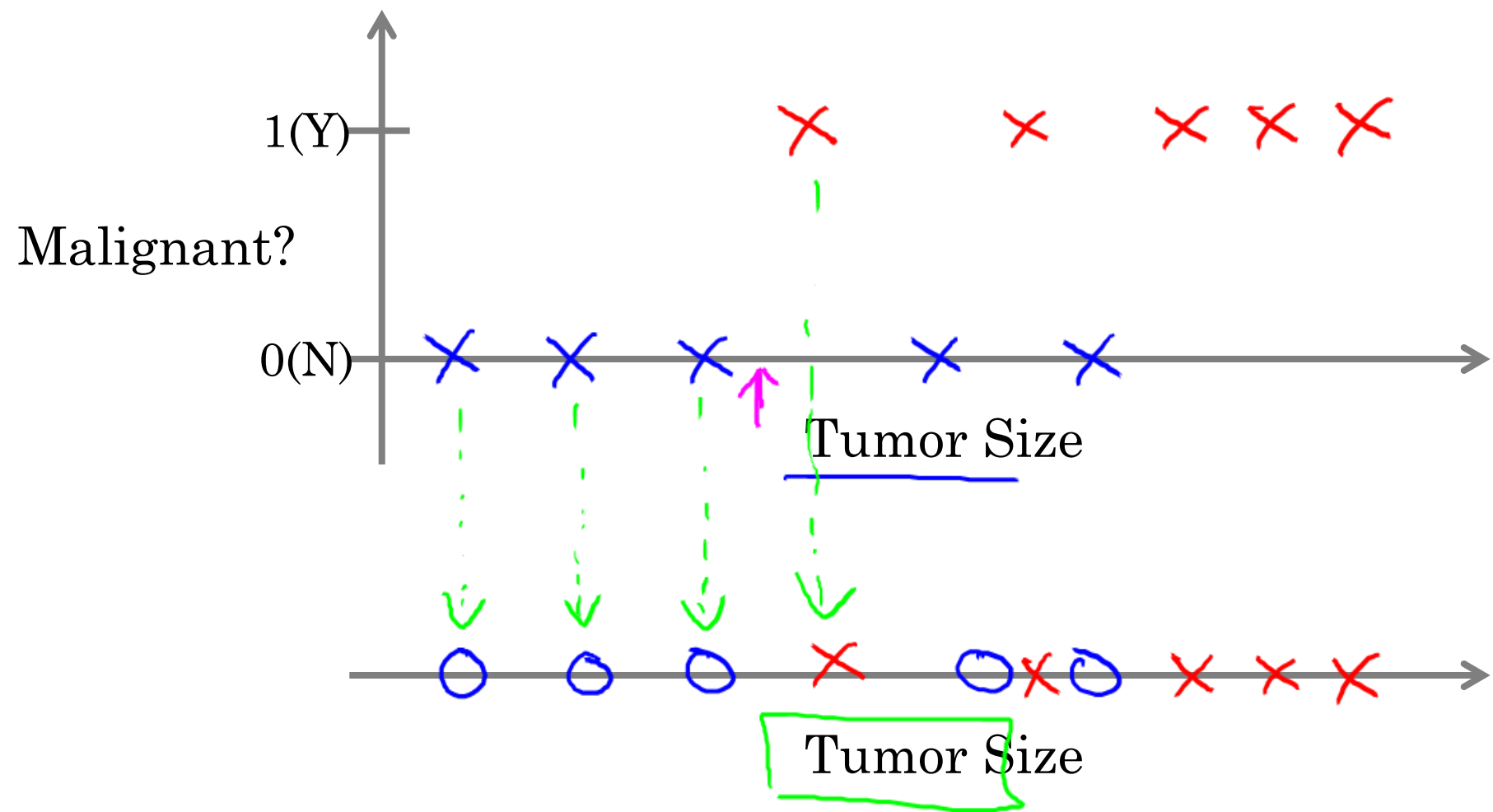
Housing price prediction



Supervised Learning
“right answers” given

Regression: Predict
continuous valued output
(price)

Breast cancer (malignant, benign)



Classification

Discrete valued
output (0 or 1)

Classification vs. Regression: Identifying the Right Approach

- You're running a company, and you want to develop learning algorithms to address each of two problems.
- **Problem 1:** You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.
- **Problem 2:** You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.
- Should you treat these as classification or as regression problems?

1. Treat both as classification problems.

2. Treat problem 1 as a classification problem, problem 2 as a regression problem.

3. Treat problem 1 as a regression problem, problem 2 as a classification problem.

4. Treat both as regression problems.

Types of Machine Learning

2. Unsupervised Learning

- Unsupervised Learning involves training a model on data that does not have labeled responses.
- The model tries to find hidden patterns or intrinsic structures in the input data.

Common Tasks in Unsupervised Learning:

1. Clustering:

- Grouping similar data points together.
- For example, customer segmentation in marketing, where customers are grouped based on purchasing behavior.

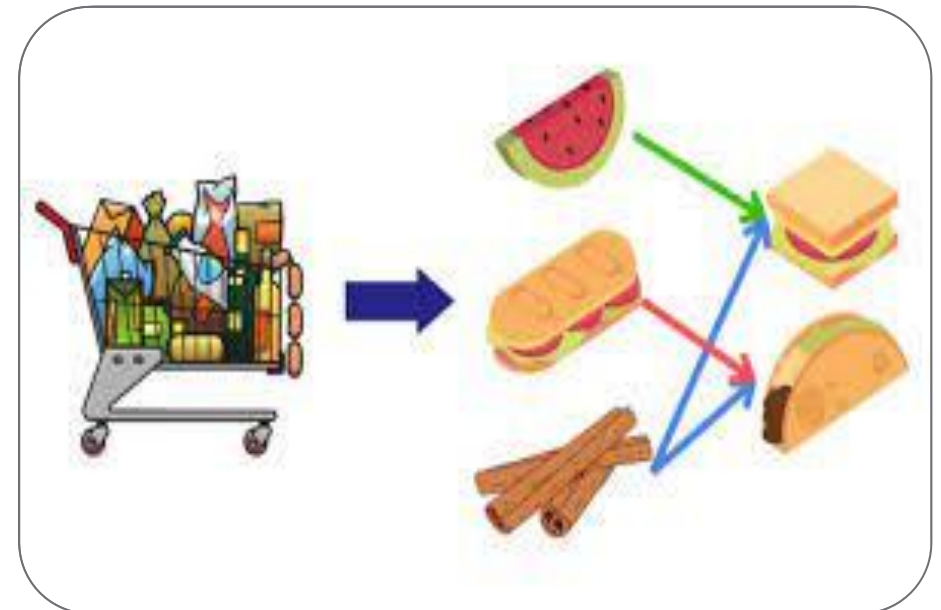
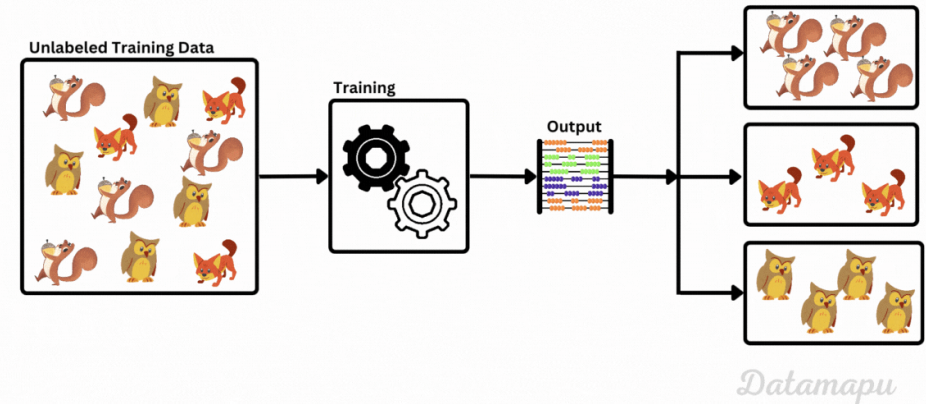
2. Anomaly Detection:

- Identifying outliers or unusual data points that do not conform to the expected pattern.
- For example, detecting fraudulent transactions in financial systems.

3. Association:

- Discovering rules that describe large portions of the data.
- For example, market basket analysis in retail, where associations between products bought together are identified.

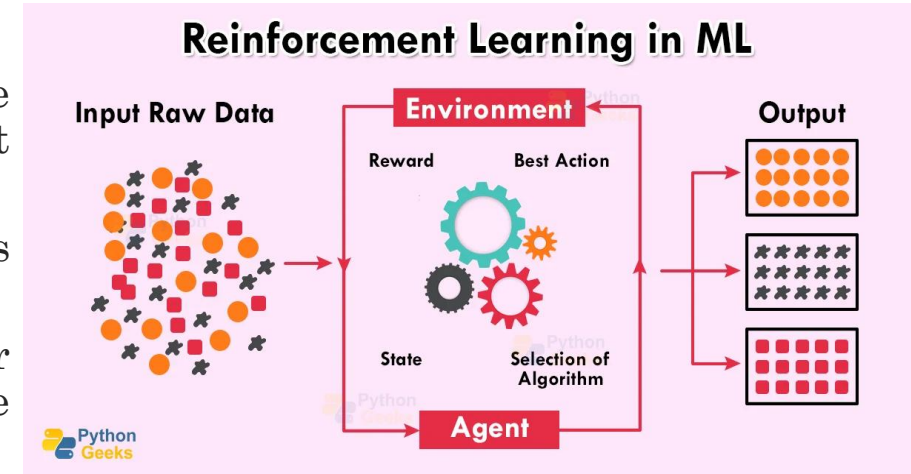
Unsupervised Learning



Types of Machine Learning

3. Reinforcement Learning

- Reinforcement Learning is a type of machine learning where an agent learns by interacting with an environment, not from fixed labeled data, but from trial and error.
- It involves training a model to make sequences of decisions by interacting with an environment.
- The model receives feedback in the form of rewards or penalties and learns to optimize its actions to maximize cumulative rewards.



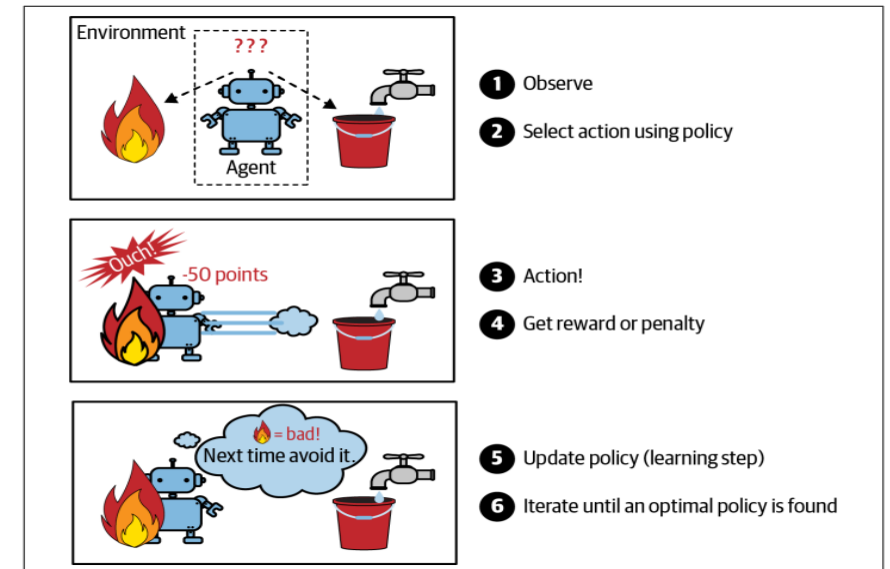
Common Reinforcement Learning Tasks:

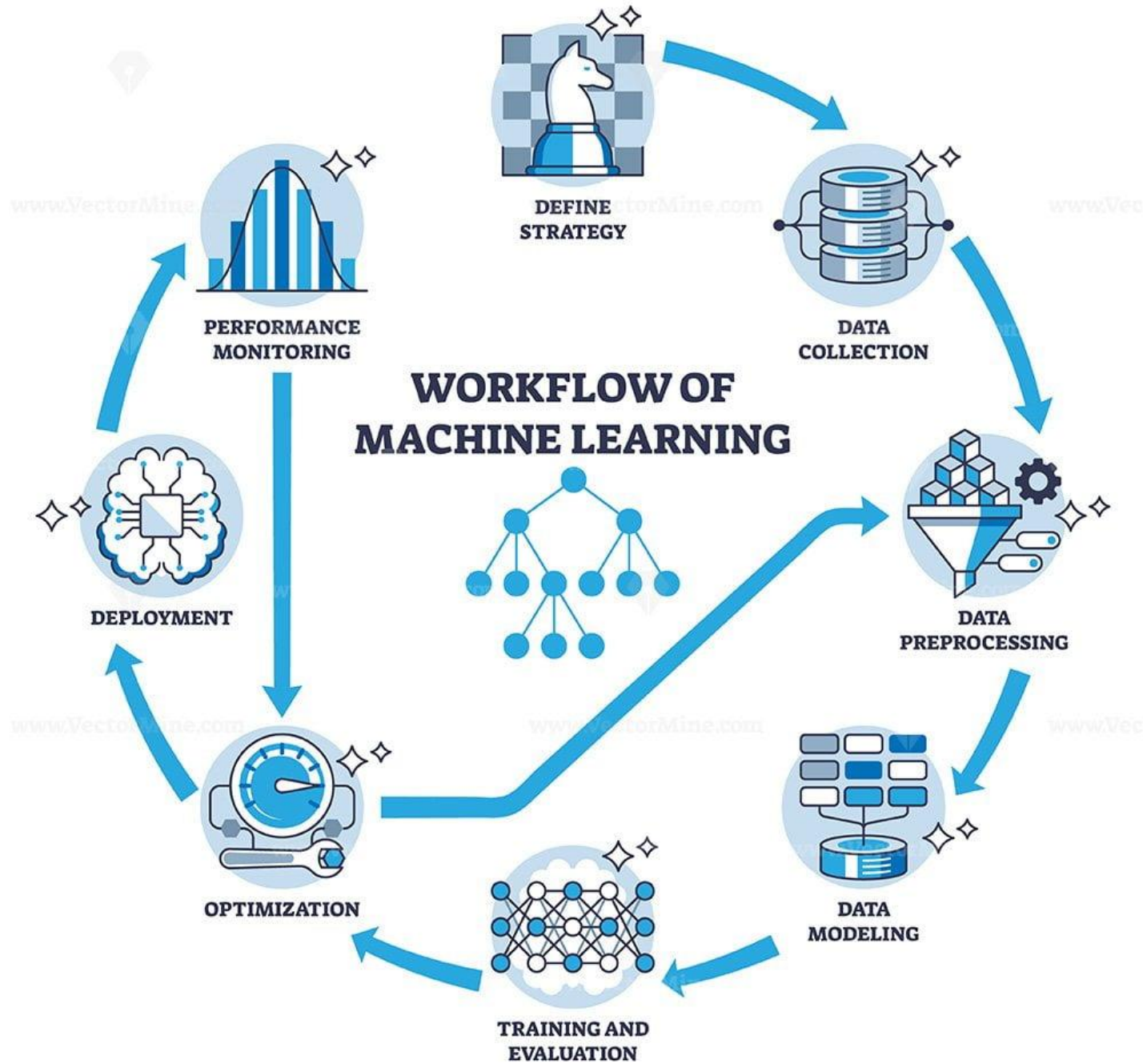
1. Game Playing:

- Training an AI to play games like Chess or Go, where the AI learns strategies through trial and error and feedback from wins or losses.

2. Robotics:

- Teaching robot to perform tasks such as walking, picking up objects, or navigating spaces, based on feedback from the environment.





Key Concepts

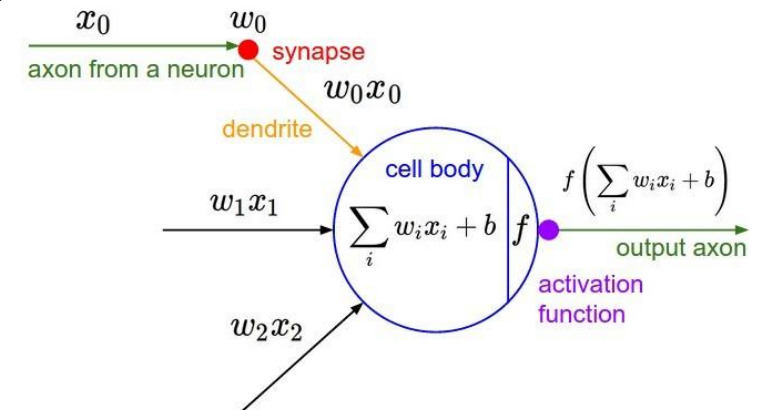
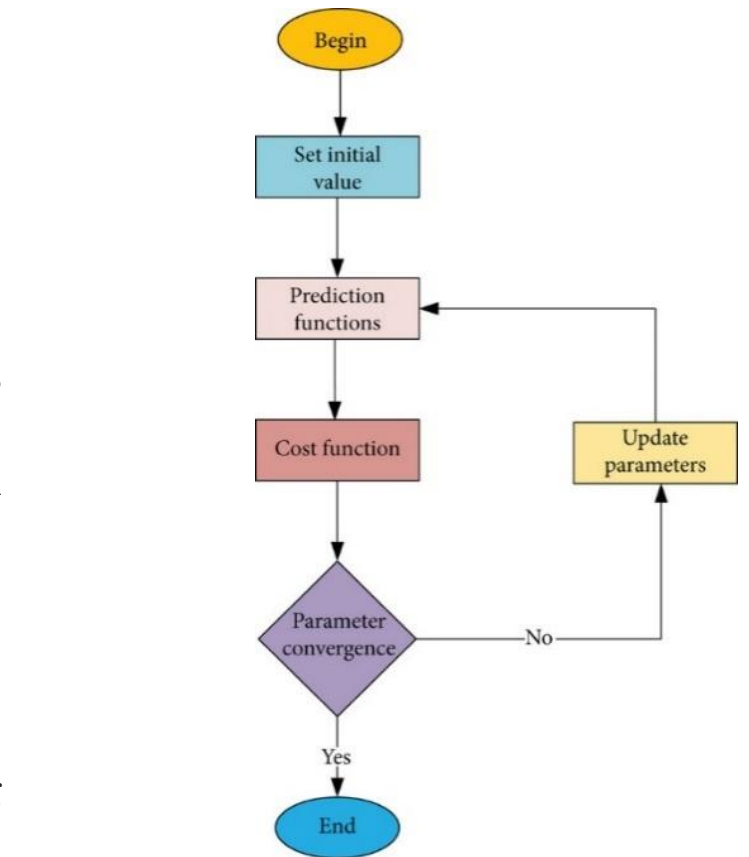
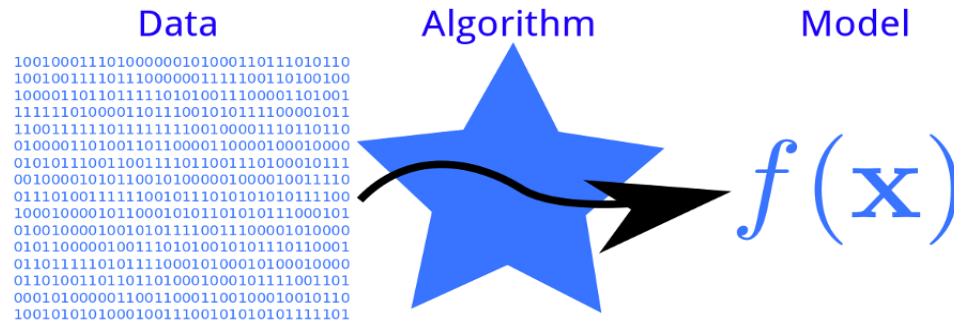
Algorithms and Models

• Algorithms

- Step-by-step procedures or mathematical formulas used to process data and learn patterns.
- In machine learning, algorithms are the methods used to train models on data.
- **Example:** Linear Regression, Logistic Regression, Neural Networks.

• Model

- The final outcome of a machine learning algorithm after being trained on data.
- A models can make predictions or decisions when given new data.
- **Example:** A trained model that predict house prices based on features like size, number of bedrooms, and location.



Key Concepts

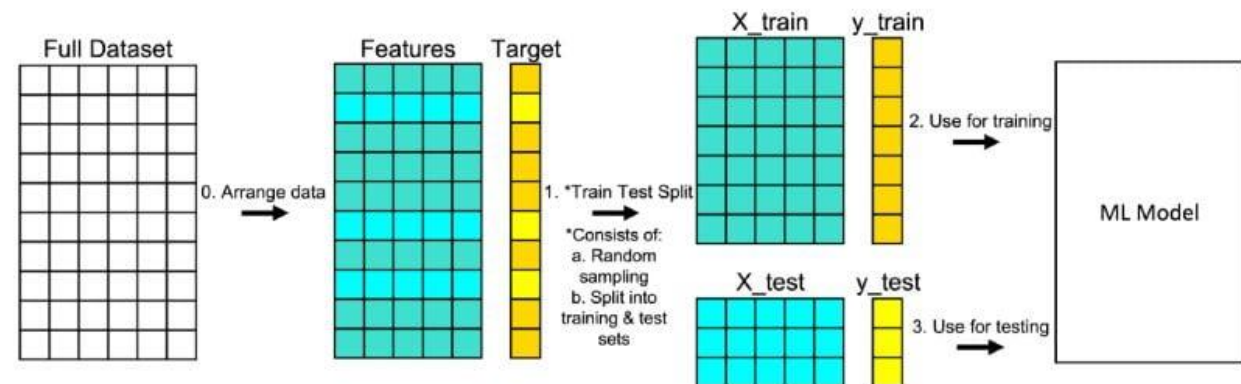
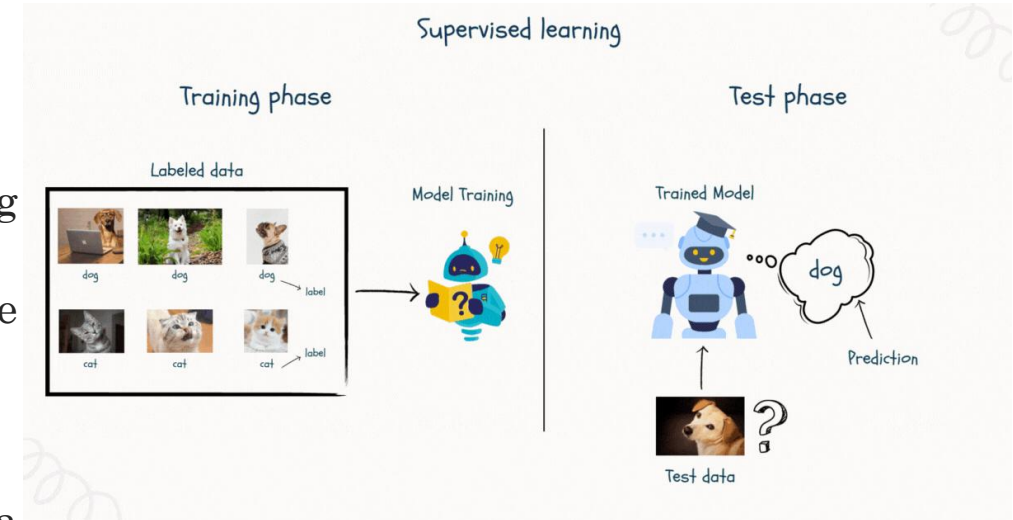
Training and Testing

- **Training**

- The process of teaching a machine learning model using a dataset.
- The model learns patterns and relationships from the training data.

- **Testing**

- The process of evaluating the trained model on a separate dataset (not seen during training).
- Helps assess how well the model performs on new, unseen data and whether it generalizes effectively.



Key Concepts

Evaluation Metrics

- Evaluation metrics are used to assess the performance of machine learning models.
- Different tasks require different metrics, depending on the type of prediction problem.

For Classification Tasks:

- **Accuracy**

- The proportion of correct predictions out of all predictions made.

- **Precision**

- The proportion of correctly predicted positive cases out of all predicted positives.

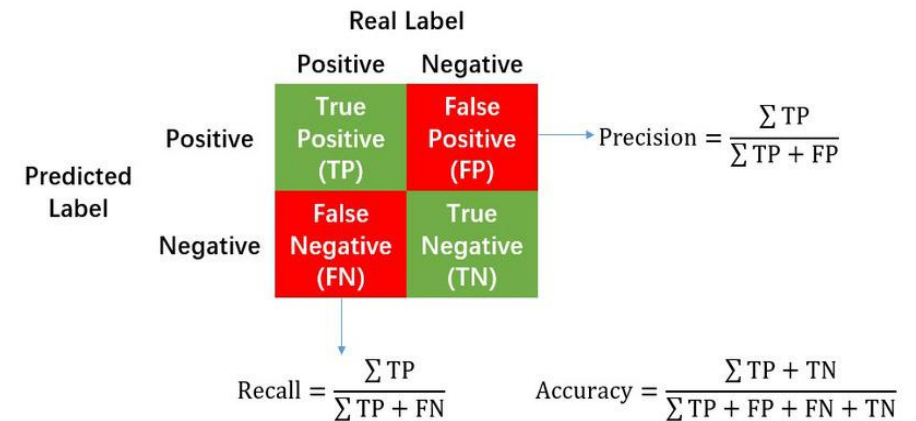
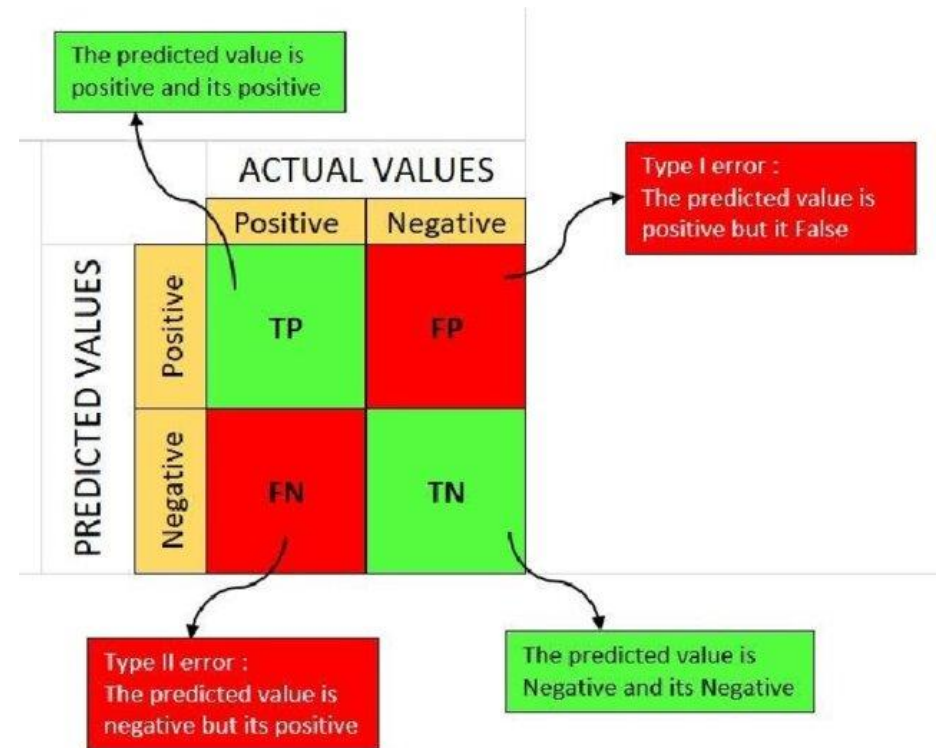
- **Recall**

- The proportion of correctly predicted positive cases out of all actual positives.

- **F1 Score**

- The harmonic mean of precision and recall — useful when class imbalance exists.

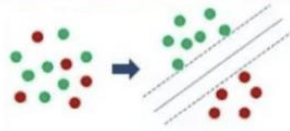
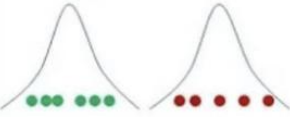
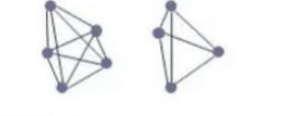
$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$



Common Algorithms

Supervised Learning

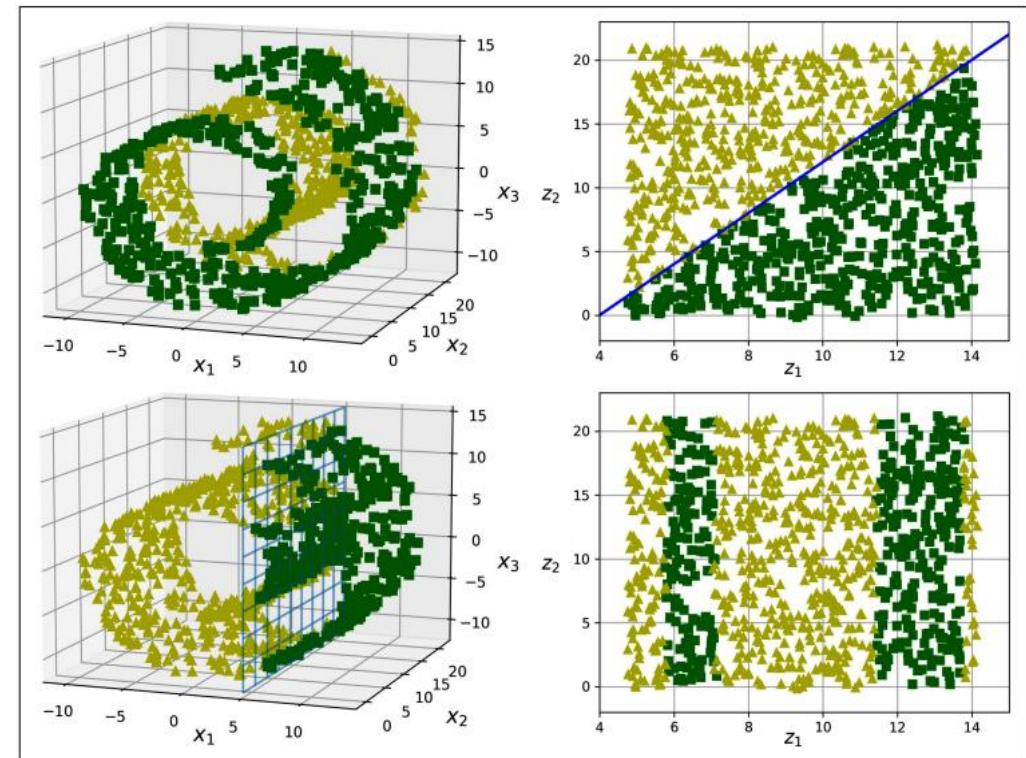
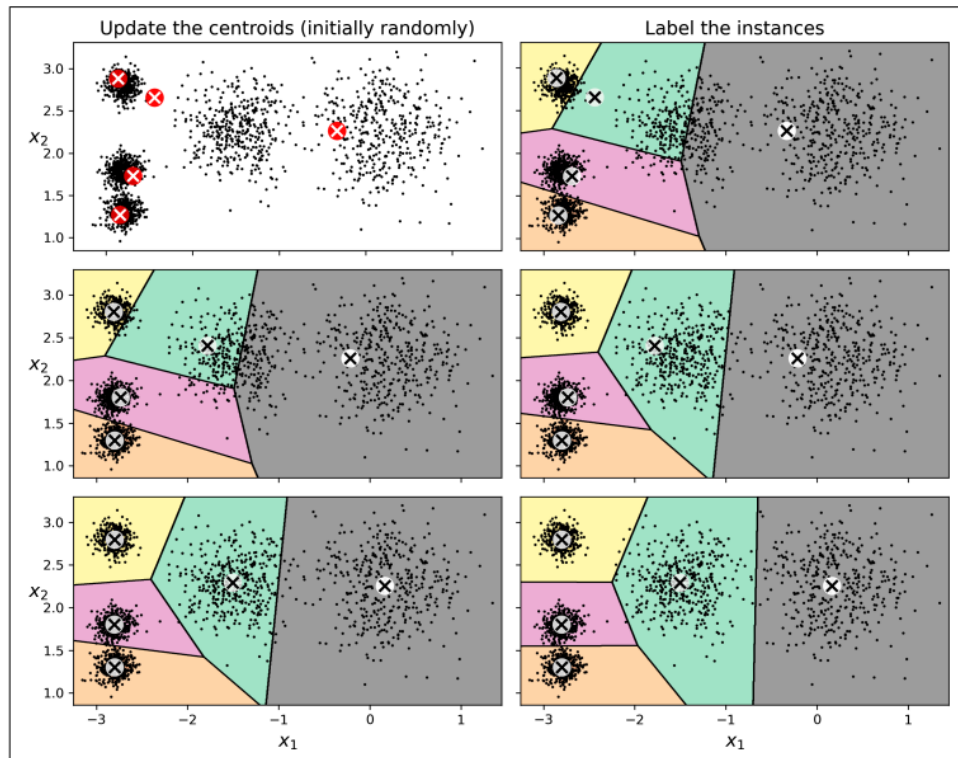
1. **Linear Regression:** Predicting continuous values.
2. **Logistic Regression:** Predicting discrete values.
3. **Decision Trees:** Making decisions based on data splits.
4. **Support Vector Machines (SVM):** Classifying data points.
5. **Neural Networks:** Mimicking brain functions for complex tasks.

Algorithm	Keyword	Diagram
Support Vector Machines (SVM)	Vector on Points	
Naïve Bayes	Probability Distribution	
Linear Regression Logistic Regression	Straight Line Logarithmic Line	
K-Means	Kernel (<i>central</i>) Mean	
K-Nearest Neighbour	Neighbouring Points	
Decision Trees	Tree Branches	
Neural Networks	Network with Layers of elements	

Common Algorithms

Unsupervised Learning

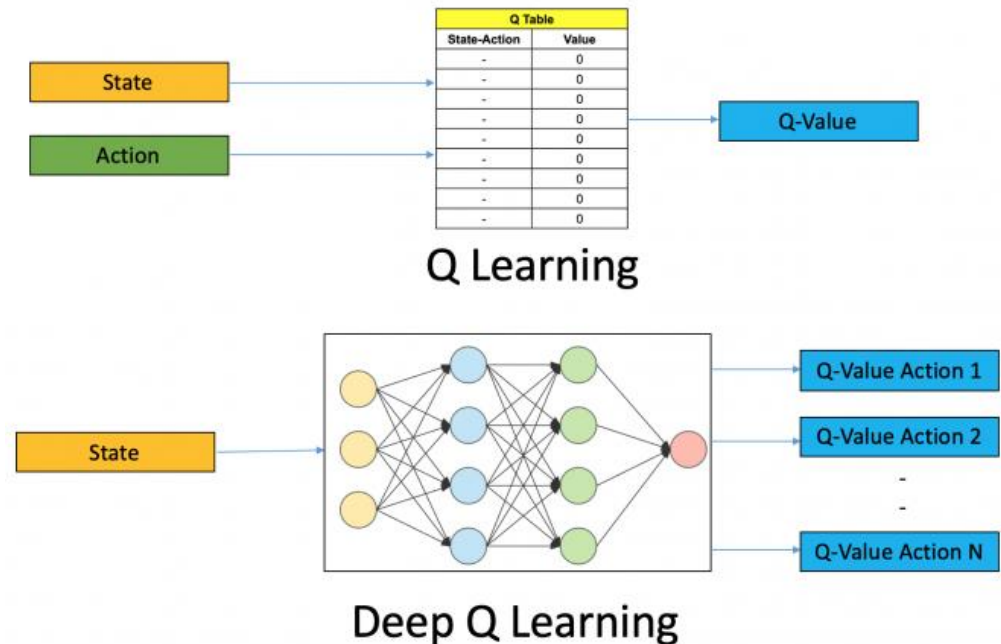
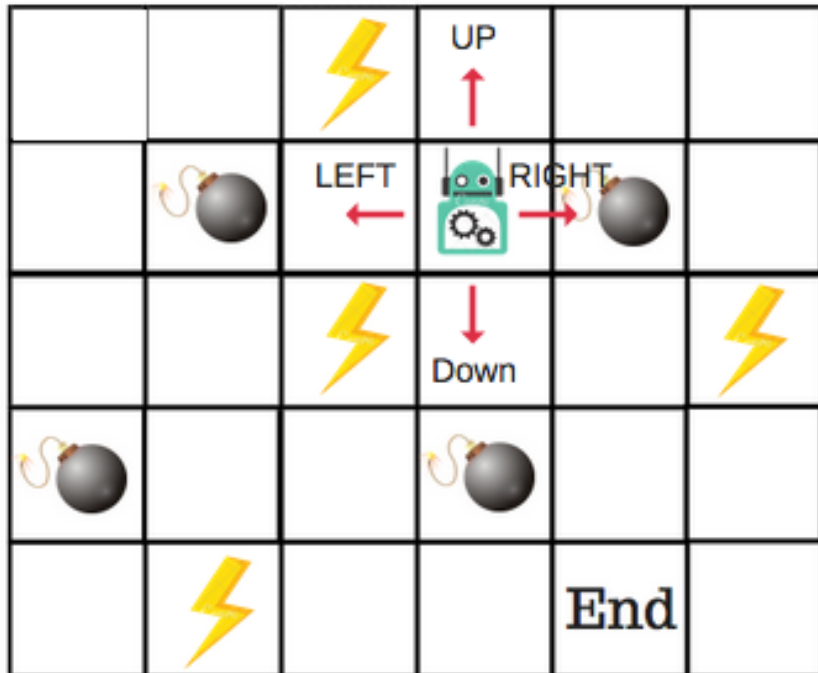
1. **K-Means Clustering:** Grouping similar data points.
2. **Principal Component Analysis (PCA):** Reducing dimensionality.



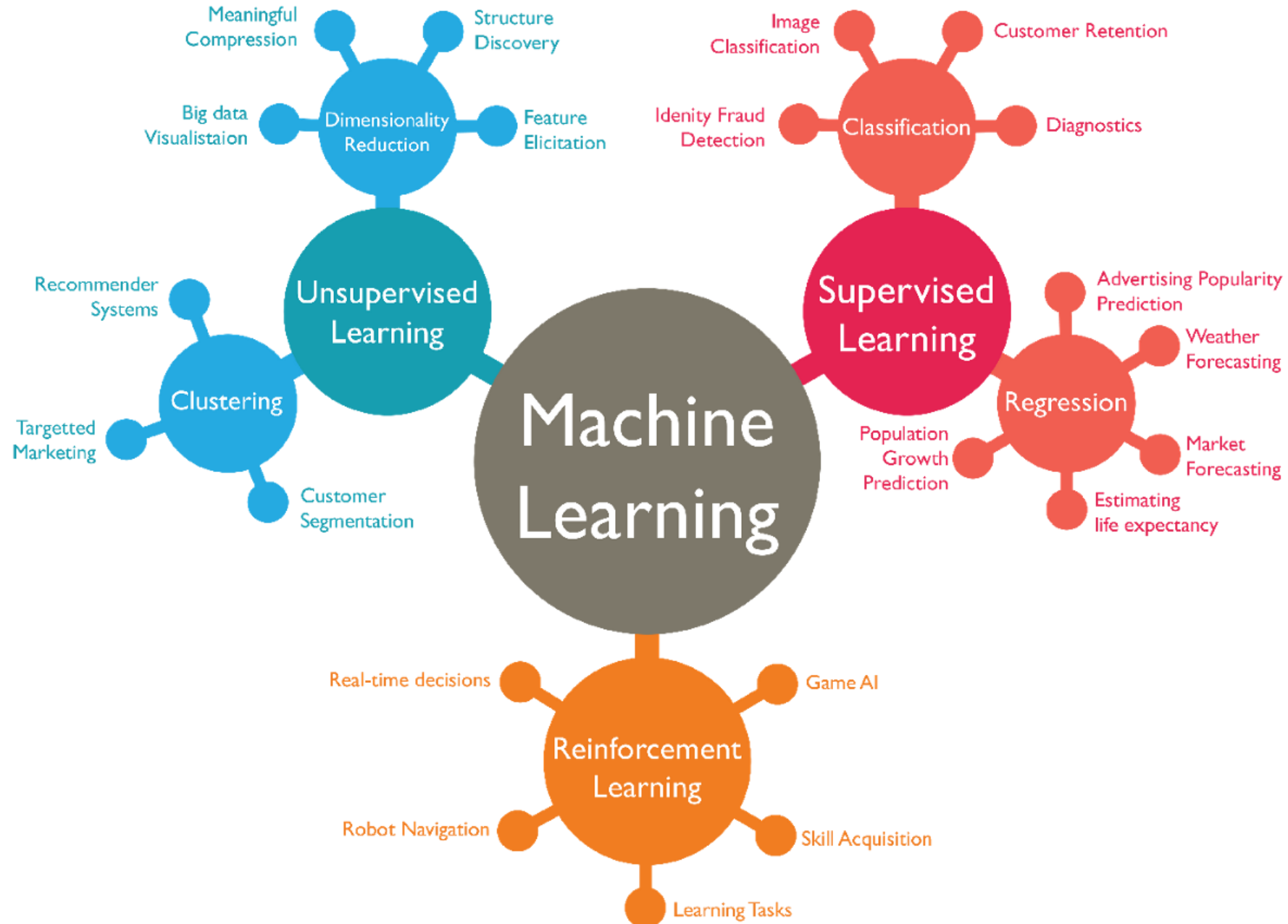
Common Algorithms

Reinforcement Learning

1. **Q-Learning:** Learning optimal actions.
2. **Deep Q-Networks (DQN):** Advanced game-playing models.



Types and Applications of Machine Learning



Tools and Frameworks

- **Programming Languages**

- **Python:**

- Most widely used language in ML.
 - Known for its simplicity and rich ecosystem.

- **R:**

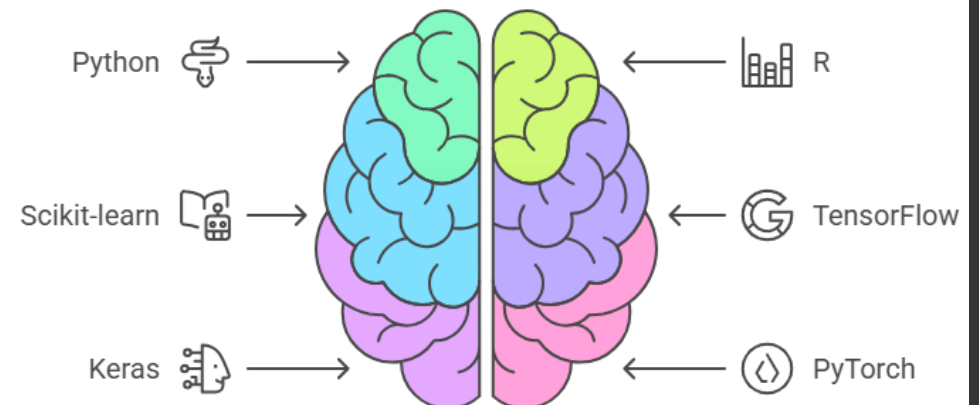
- Preferred for statistical analysis and data visualization, especially in academic and research settings.

- **Key Libraries and Frameworks**

- **Scikit-learn:** A comprehensive library for classical machine learning algorithms (e.g., regression, classification, clustering), ideal for beginners and rapid development.
 - **TensorFlow:** A powerful open-source library developed by Google for building and deploying machine learning and deep learning models.
 - **Keras:** A high-level API running on top of TensorFlow; great for quick prototyping and building neural networks.
 - **PyTorch:** A flexible and widely adopted deep learning framework developed by Facebook; popular in both research and production.

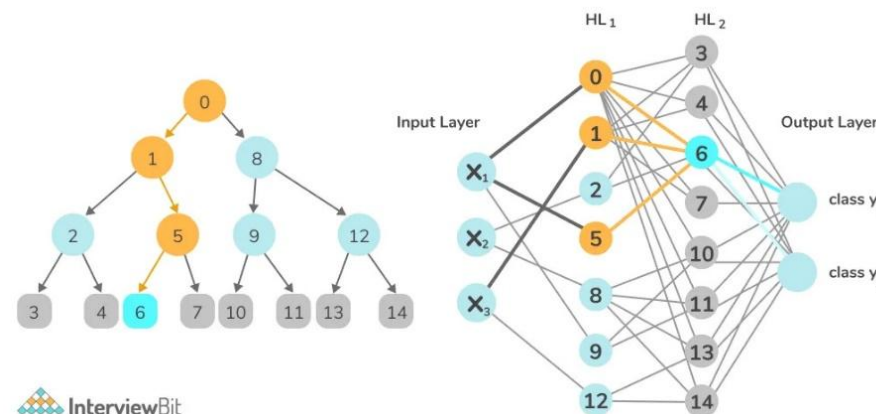
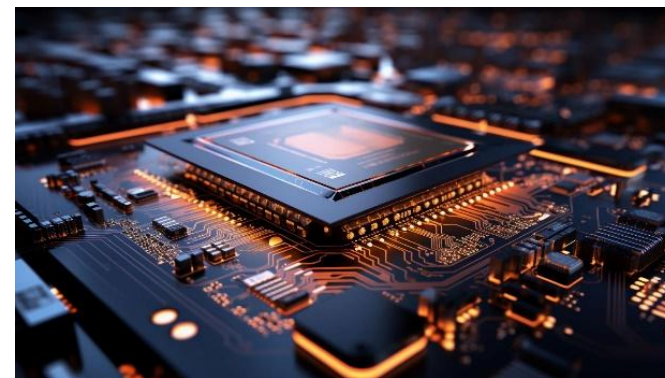
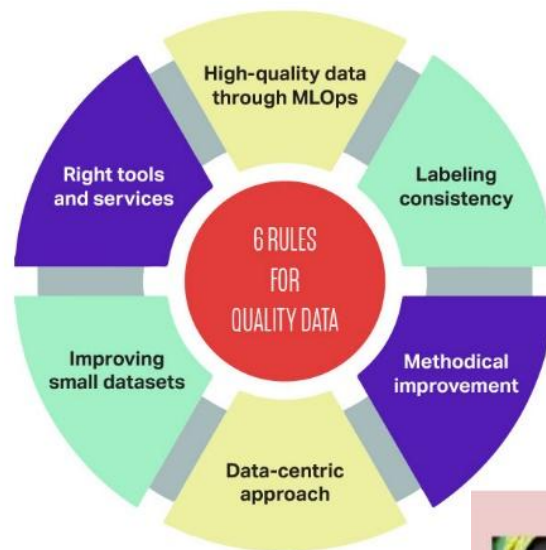


Machine Learning Tools and Frameworks



Challenges and Limitations

- **Data Quality and Quantity**
 - Need for large, high-quality datasets.
- **Computational Power**
 - High resource requirements.
- **Model Interpretability**
 - Difficulty in understanding complex models.
- **Ethical and Privacy Concerns**
 - Issues around data usage and bias.



Emerging and Future Trends

- **AutoML**

- Automates the process of building, training, and tuning ML models, making ML more accessible to non-experts.

- **Edge Computing**

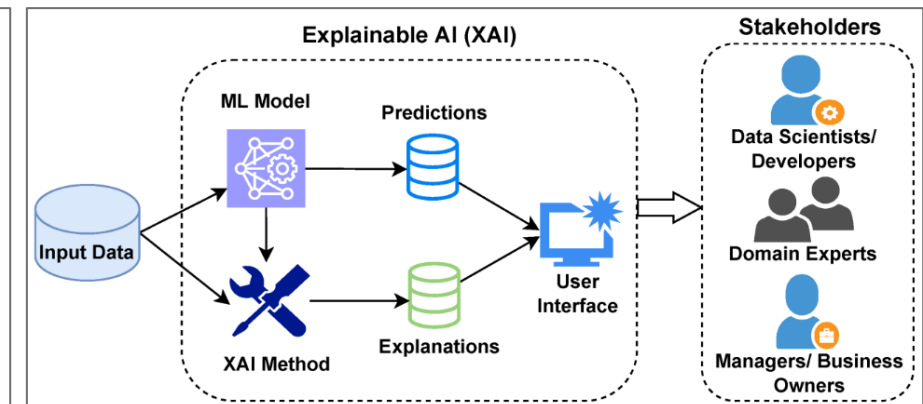
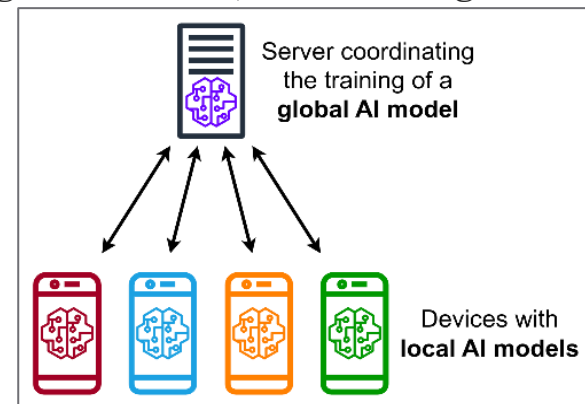
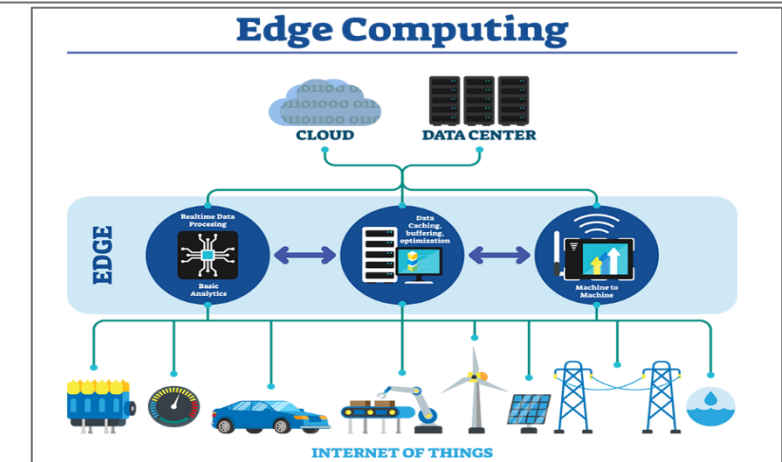
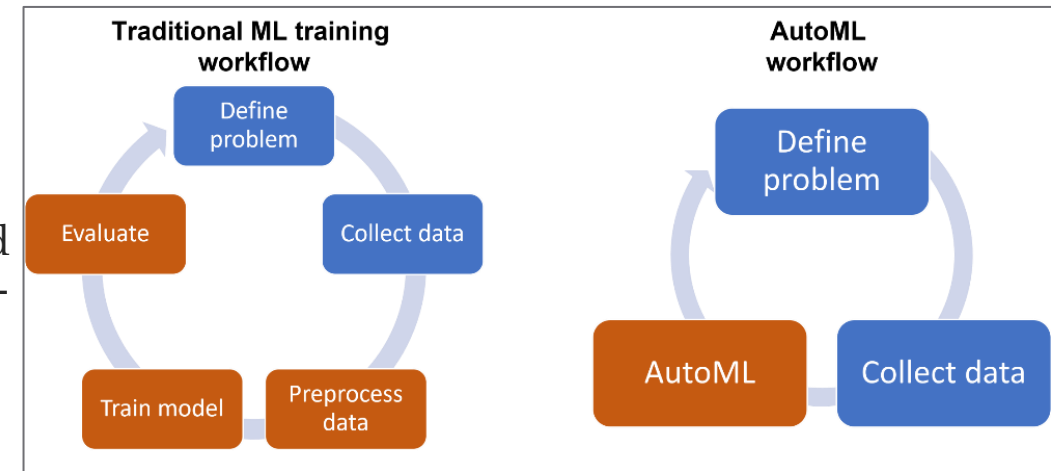
- Runs ML models directly on devices like smartphones, sensors, and IoT devices, reducing latency and dependence on the cloud.

- **Explainable AI (XAI)**

- Focuses on making ML models more transparent and interpretable, helping users understand how decisions are made.

- **Federated Learning**

- Enables training models across multiple devices or organizations without sharing raw data, enhancing privacy and data security.



Thank You